FOR ENGINEERING, CONTROL SYSTEMS AND IT TEAMS

Whatever runs your network, it already speaks a language FlowSense reads.

FlowSense is a hosted platform for sewer network monitoring, flow analytics and hydraulic modelling. Getting your readings into it is not a project. There are five ways in, each of them outbound from one side only, and every SCADA package, RTU fleet and equipment supplier's cloud we have met supports at least one. This brochure is written to be forwarded: the first half is for whoever will make the connection, in enough detail to build from, and the second half shows what the readings become once they arrive.

FIVE WAYS IN

Your SCADA posts to us over HTTPS, we subscribe to your MQTT broker, we poll your API or your supplier's cloud, we collect CSV exports from your server, or you upload them to a folder we host.

WHAT YOU INSTALL

Nothing. No agent, no client, no gateway appliance and no browser extension, at your sites or in your control room.

WHAT YOU OPEN INBOUND

Nothing at all if your SCADA posts to us or sends its files to the folder we host. For the other three, one service you already choose to expose, restricted to a single EDS source address.

WHAT WE NEVER TOUCH

OPC UA, DNP3, Modbus, EtherNet/IP and S7 are never polled over the internet, at any stage. No VPN, no jump host, no route into your control network.

TIME TO FIRST READING

An HTTPS push can be sending inside an hour of the token being issued, because there is no firewall change in front of it, and a file drop needs one outbound rule at most. The other three wait on your change process, not on us.

WHERE IT ENDS UP

Validated readings on named channels, feeding inflow and infiltration analysis, hydraulic models, pump station analytics, alarms and reports. Yours to export at any time.

1. THE SHORT VERSION

Five ways in, one way through.

The five options differ only in who opens the connection and what carries the readings. After that they are the same path: one validated ingest core, one review step where a person decides what each of your tags means, and then ordinary data with the same charts, alarms, models and exports as everything else in the platform.

Your instruments
Flow and level meters, rain gauges, pump station PLCs, water quality sensors.

THEN
your network

Your SCADA, historian or supplier's cloud
Where those readings already land today. Nothing here changes.

ONE OF FIVE
outbound

FlowSense ingest
Timestamps checked, values read, duplicates dropped, unknown tags held for review.

THEN
your account

Channels and analysis
Named channels on your sites, feeding every tool in the platform.

The three things worth knowing before you read any further

- **Your SCADA cannot create anything in FlowSense.** A tag the platform has not seen before is held in a review queue with a sample value. Its readings are held aside, never written to a channel, until a person points that tag at one. Nothing your system sends can invent a site, a channel or a measurement type.
- **A FlowSense credential cannot read anything in yours.** The token your integrator is given is a separate credential class that can only write, and only into its own connection. Where we connect to you instead, we ask for a read-only account scoped to one folder, one topic tree or one API.
- **Every transport is safe to re-send.** Readings are deduplicated on the channel and the instrument's own timestamp, so a retry, a replayed file or an overlapping window costs duplicates dropped, never duplicates stored. Nobody has to build exactly-once delivery to talk to us.

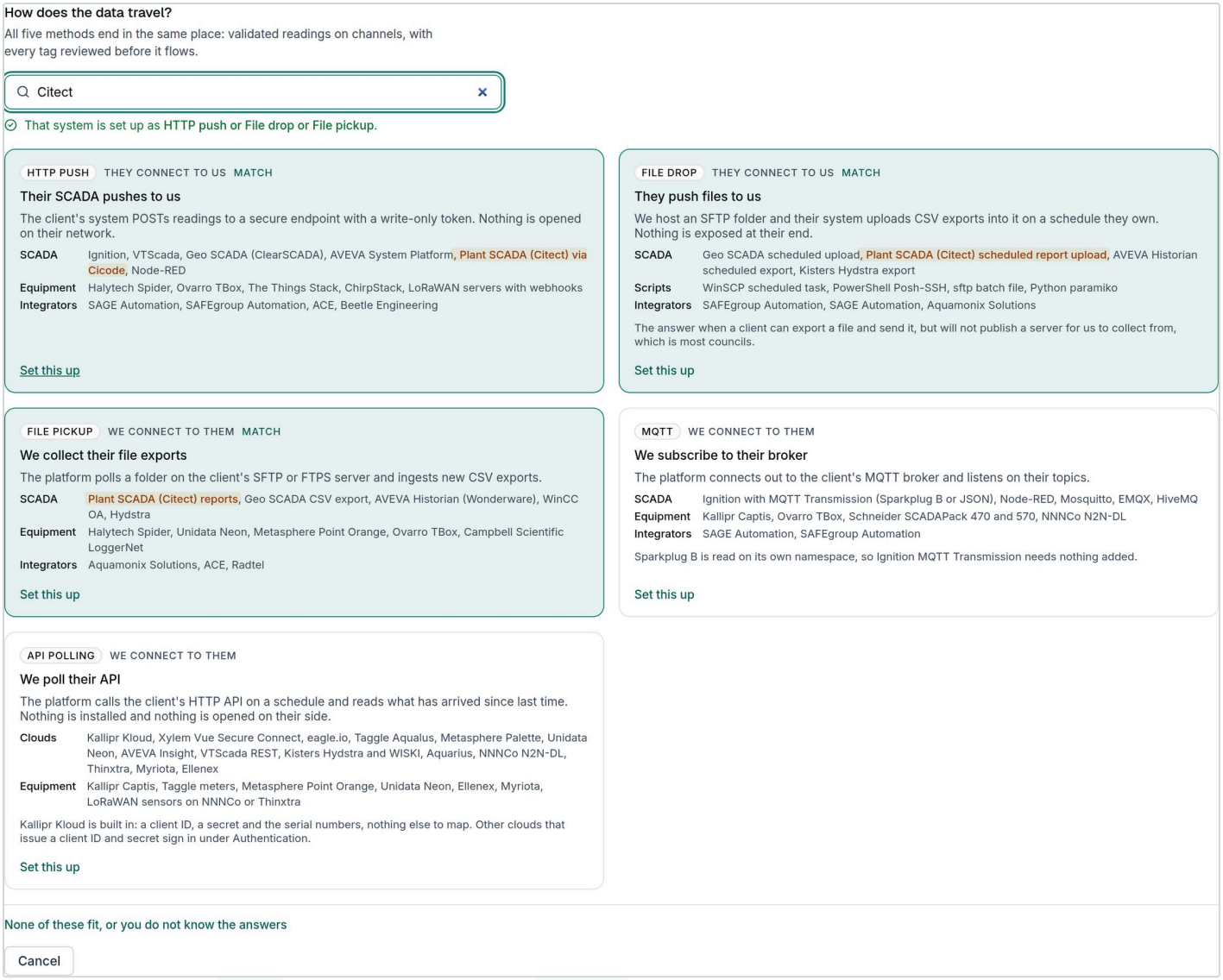
2. FIND YOUR SYSTEM

You do not have to know what a transport is. Name what you run.

The platform's connection screen is built around the question a council actually asks, which is "we run Citect, now what". Type the name of the package in your control room, the RTUs in your pits or the cloud your loggers report to, and it answers with the method that fits and the form to fill in. The table below is the same answer on paper.

WHAT YOU RUN, AND HOW IT REACHES FLOWSENSE

IF THIS IS WHAT YOU HAVE	THE WAY IN	WHO OPENS THE CONNECTION
Ignition, VTScada, Geo SCADA (ClearSCADA), AVEVA System Platform, Plant SCADA (Citect) via Cicode, Node-RED	HTTPS push, from a script, gateway module or driver you already have	You connect out to us. Nothing inbound.
Ignition MQTT Transmission, Node-RED, Mosquitto, EMQX, HiveMQ	MQTT subscription to your broker	We connect out to you.
Kallipr Kloud, Xylem Vue Secure Connect, eagle.io, Taggle Aqualus, Metasphere Palette, Unidata Neon, AVEVA Insight, VTScada REST, Kisters Hydstra and WISKI, Aquarius, NNNCo, Thinxtra, Myriota, Ellenex	API polling, on a schedule you choose	We connect out to you or to your supplier's cloud.
Plant SCADA (Citect) reports, Geo SCADA CSV export, AVEVA Historian (Wonderware), WinCC OA, Hydstra, Campbell Scientific LoggerNet	File pickup from your SFTP or FTPS server	We connect out to you.
Geo SCADA scheduled upload, Plant SCADA (Citect) scheduled report upload, AVEVA Historian scheduled export, Kisters Hydstra export, or a scheduled script: WinSCP, PowerShell Posh-SSH, an sftp batch file, Python paramiko	File drop: the same CSV exports, uploaded to an SFTP folder we host	You connect out to us. Nothing inbound.
Halytech Spider, Ovarro TBox, Kallipr Captis, Schneider SCADAPack 470 and 570, Metasphere Point Orange, Unidata Neon, The Things Stack, ChirpStack	Push, MQTT or API polling, depending on how the fleet is configured today	Either, and we will tell you which is least work.
Ovarro Kingfisher, Schneider SCADAPack, ELPRO, Abbey Systems, or anything reached over OPC UA, DNP3 or Modbus	These report to your control room, not to us. Your SCADA then chooses one of the five above. Plant protocols are deliberately never polled over the internet. No council exposes them, and every system that speaks them supports one of the five options.	Whichever your SCADA supports.



The connection screen, answering a vendor name. Each of the five cards names the SCADA packages, field equipment, supplier clouds, scripts and integrators it fits, so the method is chosen from what the council runs rather than from a protocol glossary. If none of them fits, the same screen sends a form to your IT or SCADA team to fill in without a login.

3. THE FIVE OPTIONS, IN THE DETAIL AN INTEGRATOR NEEDS

Each one answers the same four questions.

Who opens the connection, what you expose, what we need from you, and what the platform does when something goes wrong. If you are scoping the work, this section is the scope.

OPTION ONE — Your SCADA posts readings to us

You connect to us, over HTTPS on TCP 443

The lowest friction of the five, and the one to choose if your package can make an HTTPS request at all. There is no firewall change, because the connection is outbound from your SCADA or historian host, the same direction as any web request. Ignition, VTScada, Geo SCADA, AVEVA System Platform and Plant SCADA via Cicode can all do this with a script.

What we give you	An endpoint and a write-only ingest token, shown once and stored only as a hash. The token carries the connection's identity, so there is no username, no site list and no user account behind it. It cannot read a single stored value.
What you send	A JSON batch of readings, each with your own tag name, the instrument's timestamp and a value. Up to **10,000 readings per request** and **240 requests per minute**, which is over two million readings a minute and well past what any council network produces. Beyond it the answer is a 429 with a retry-after header rather than a dropped batch.

What comes back	A count of what happened to every row: inserted, duplicate, rejected, awaiting mapping, skipped. A malformed reading is **one rejected row with its position in the batch** and the reason, never a refused batch: the other 9,999 readings still land.
If we are unreachable	Send the same batch again when we are back. Duplicates are dropped on the channel and timestamp, so a client that never heard our answer loses nothing by repeating itself.

THE FIRST CALL AN INTEGRATOR SHOULD MAKE

```
GET https://edsflowsense.au/api/v1/ingest/v1/whoami
Authorization: Bearer edsw_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

{
  "connection": "Council SCADA (Ignition)",
  "timezone": "Australia/Brisbane",
  "timestamp_mode": "field",
  "tags": { "mapped": 24, "unmapped": 3, "ignored": 0 },
  "server_time": "2026-09-14T07:20:11.482Z",
  "max_readings_per_request": 10000,
  "rate_limit_per_minute": 240
}
```

It confirms the token, names the connection it feeds, and returns our clock. Clock skew is the single most common reason a first push is refused, so it is worth comparing before you send anything.

SENDING READINGS

```
POST https://edsflowsense.au/api/v1/ingest/v1/readings
Authorization: Bearer edsw_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Content-Type: application/json

{
  "readings": [
    {"tag": "PS02.WETWELL_LEVEL",  "timestamp": "2026-09-14T07:15:00+10:00", "value": 1.442, "units": "m"},
    {"tag": "PS02.DISCHARGE_FLOW", "timestamp": "2026-09-14T07:15:00+10:00", "value": 31.3, "units": "L/s"},
    {"tag": "PS02.PUMP1_RUN",      "timestamp": "2026-09-14T07:15:00+10:00", "value": 1},
    {"tag": "STP.INLET.FLOW",      "timestamp": 1757805300,                  "value": 140.4}
  ]
}
```

AND WHAT THE PLATFORM SAYS BACK

```
{
  "ok": true,
  "received": 4, "inserted": 3, "duplicates": 0, "rejected": 0,
  "unmapped": 1, "ignored": 0, "skipped": 0,
  "counter_baselines": 0, "counter_resets": 0,
  "quarantined": 1, "tags_discovered": 1,
  "reject_details": [], "problems": [], "notes": [],
  "note": "1 new tag(s) recorded for mapping. Their readings are held back until
           an operator maps them to channels in SCADA connections"
}
```

What the fields mean, and why there are so many of them

- **inserted** and **duplicates** are the normal outcome. A duplicate is not an error: it is the dedupe doing its job on a resent window.
- **rejected** is a row the platform could not honestly store, and `reject_details` names each one by its position and reason, up to fifty per response. The rest of the batch is unaffected.
- **unmapped** and **quarantined** mean a tag nobody has pointed at a channel yet. Its readings are held for seven days and **replayed automatically when the tag is mapped**, so mapping a tag late still brings its history with it.
- **skipped** is a reading whose value was a quality word rather than a number: Bad, N/A, Not Connected, a row of hashes. Those are skipped, never stored as zero, because a zero in a flow series is a measurement and a fault is not.

OPTION TWO

We subscribe to your MQTT broker

We connect to you, on TCP 8883 with TLS, or 1883

The natural fit where the SCADA stack already publishes to a broker, which is how the MQTT era expects to be consumed. The platform connects out to your broker as an ordinary subscriber and listens on the topics you nominate. You expose a broker and a user limited to those topics, restricted to our source address; you do not expose the SCADA behind it.

What we need	Hostname and port, the topic filters (wildcards are fine, for example `plant/+/flow` or `site1/#`), a broker user, and the certificate authority if your TLS is issued internally. An administrator can fetch your server's certificate fingerprint through the platform, confirm it with you out of band, and pin it, so a substituted broker cannot collect the password.
Payloads we read	A bare number on a topic, a JSON object with the value and timestamp at paths you name, an array of such objects, or an object carrying several numeric fields, each read as its own tag. **Sparkplug B is decoded directly on the `spBv1.0/` namespace**, so Ignition MQTT Transmission needs no second JSON namespace and no script: metric names are learned from the birth certificates, and an alias with no name behind it yet is reported rather than guessed at.
Our downtime is not your data loss	The subscription is a **persistent session under a stable client identifier**, so your broker keeps the subscription and queues QoS 1 messages while we deploy or restart, then replays them when we return. Publish at QoS 1: no broker queues QoS 0, so a restart would miss it, and that is a property of MQTT rather than of us.
If it cannot connect	The failure is written to the connection with the reason and the host and port it tried. If the port answers like a plant protocol, the platform names that in the error: losing a connection to a broker and pointing at a Modbus port look identical otherwise.

OPTION THREE

We poll your API, or your equipment supplier's cloud

We connect to you, over HTTPS on TCP 443

The option for remote equipment with no control room behind it, which is most fleets arriving without EDS loggers: the gear reports to a vendor cloud, and that cloud has a REST API. It is also the option for a council whose own systems expose an API but cannot push. Nothing is installed and nothing is opened on your side, because a supplier's cloud is already public.

Configured, not coded	A URL with placeholders for the window being asked for, the authentication style (a bearer token, a key in a header or the query string, a username and password, or a client ID and secret exchanged for a token under OAuth2 client credentials and renewed before it expires), whether the response is JSON or CSV, whether records carry a tag each or a column per tag, and the field names. A paged response is followed by the path to the next page, up to twenty pages per poll.
Vendor clouds we hold the protocol for	**Kallipr Kloud is built in.** It needs a client identifier, a secret and the serial numbers, and nothing else to map: the token exchange, the paging, the per device requests and the overlap are all the platform's problem. Alarms are never read, only sensor and health data.
We ask for a window, never for everything	Each poll asks from a watermark to now, overlapping by five minutes so an API that reports late does not drop rows into a gap, and capped at seven days. The watermark advances **only when everything received was actually dealt with**, so a window whose tags are still unmapped is asked for again and mapping the tag brings its history with it.

A fleet keeps a position per logger	Where one credential serves many devices, the platform holds a watermark per serial as well as for the connection. Otherwise a logger that goes quiet for a week and then uploads its backlog is never asked for the older half of it, and the hole is invisible because its recent readings do arrive.
What we refuse to fetch	Plain HTTP, loopback, private and link local ranges, and cloud metadata addresses, checked before the name is resolved and again on every page. Redirects are not followed. An administration form that makes a server fetch a URL is a request forgery tool unless it refuses to point inwards.

OPTION FOUR

We collect the CSV exports your historian already writes

We connect to you, SFTP on TCP 22 or FTPS on TCP 21

The lowest common denominator, and for that reason the most important option for older council SCADA. If your historian already writes a report every fifteen minutes, or overnight, the whole integration is a folder and an account confined to it. Nothing about the export has to change.

What we need	Hostname, the folder, a filename pattern such as `*.csv`, and a read-only account scoped to that folder. SSH keys are preferred to passwords and we will send you our public key. The server's host key or TLS certificate can be pinned after you confirm the fingerprint out of band.
What the reader tolerates	Banner lines above the header, a units row under it, quality columns beside each value, separate date and time columns, semicolons or tabs, UTF-16 byte order marks, thousands separators, decimal commas, values with their units attached such as `42.5 L/s`, and digital states written as ON, OFF or RUNNING. Real historian exports are messy, and refusing them is not a standard, it is a support call.
What it refuses to guess	**Ambiguous dates.** You say whether 03/04/2026 is March or April; left on automatic, the reader accepts only files where one reading explains every row and rejects the rest with the reason. A transposed day and month is the kind of error that is invisible for eleven months of the year.
What it does with your folder	Remembers what it has read, by name, size and modification time, so a file the historian keeps appending to is noticed and re-read while an unchanged one is left alone. Up to twenty files per run, thirty megabytes and five hundred thousand rows per file. **Nothing is ever deleted unless you ask for it**, and then only from a file that landed with no rejects and no unmapped tags.

A HISTORIAN EXPORT THE READER TAKES AS IT STANDS

```
Plant SCADA Report generated 14/09/2026 07:20
Site: Pump Station 2

Date,Time,Wet Well Level,Wet Well Level.Quality,Discharge Flow,Pump 1
,,m,,L/s,
14/09/2026,07:15:00,1.442,Good,31.3,ON
14/09/2026,07:20:00,1.401,Good,30.9,ON
14/09/2026,07:25:00,1.287,Bad,,OFF
```

The banner is skipped, the units row is recognised as units rather than as a reading, the date and time columns are joined, ON and OFF become 1 and 0, and the row whose quality is Bad is **skipped rather than stored as zero.** The accommodations are recorded against the run so an engineer can see what was assumed, and none of them is reported as an error, because a file the platform read correctly is not a partial delivery.

OPTION FIVE

You send those same exports to a folder we host

You connect to us, SFTP on TCP 2222

Option four turned around, and the answer when the file is not the problem but the server is. Most utilities can schedule an outbound transfer this afternoon and cannot get approval to publish a server for a supplier to collect from, and that single fact

decides more integrations than any technical detail on this page. Here we run the server, you push, and nothing at your end is exposed.

What we give you	A hostname, a port, and a login confined to your own folder with no shell, no ability to reach anything else and nowhere else to write. Plus our host key fingerprint, which your client pins so it can tell our server from anything else claiming to be it. Send us the public half of a key pair and no password need ever live in your script.
What you need to open	Outbound TCP 2222 from the one host that sends the file. Nothing inbound, and no server of yours facing the internet. Deliberately not port 22: that is our own administration, and no client service runs on it.
What we need from you	No hostname, no source address to allowlist and no account on your systems. What we need is the export itself: send one file before the connection is switched on and a test reads it without storing anything, so the date order, time zone and columns are confirmed against your real export rather than a description of it. If you would rather hold no password at all, send the public half of a key pair.
How you send	Any filename ending `*.csv`, on whatever schedule suits you, into the folder your session opens in. There is nothing to create and nothing to tidy: files are kept for a month and then removed. Re-sending after a failed transfer is safe, because readings are de-duplicated on channel and timestamp. Upload to a temporary name and rename on completion if your client can, and a half-transferred file is never read; if it cannot, a file is not read until it has been unchanged for a minute.
If a transfer fails	It fails at your end, at the moment it runs, which is the only moment anybody is watching. A file over **thirty megabytes**, or one that would take your folder past its ceiling (500 MB unless agreed otherwise), is **refused at the write** rather than accepted and dropped later, and your client reports why. If our server cannot be reached, your task fails the same way and the next run, or a re-send, covers it. The folder never fills with a steady export: once it passes 80 per cent of its ceiling the oldest files already read are removed early, and a file not yet read is never the one removed. Eight failed sign-ins from one address in five minutes block that address for a quarter of an hour.
If nothing arrives	The connection says so. For a file drop, contact means a file arrived, not that we looked in our own folder, so a connection whose task has never run reads as waiting for data rather than healthy, and one that stops sending is marked stale after a day.
What you can ask us afterwards	Every transfer is recorded as the server saw it, including one that was cut off part way, so **"we sent it last night" has an answer that does not depend on whether the file turned into readings**. A file that arrived empty and a file that never arrived look identical everywhere else; here they do not.

Everything option four says about messy exports applies here unchanged, because it is the same reader: the banner, the units row, the quality columns, the ambiguous dates it refuses to guess. The only difference is which end holds the server.

4. WHAT HAPPENS TO A TAG NAME

Mapping a hundred and fifty tags is the real cost of a connection, so the platform reads them for you.

A pump station publishes the same dozen measurements everywhere in the country, under a different name at every council. Eighteen stations at eight tags each is a hundred and fifty decisions, and every one of them is the same decision. FlowSense reads the tag names the way an engineer reads them and proposes the answer for all of them at once, so the operator confirms rather than types.

THE SAME READING, NAMED FOUR WAYS, READ CORRECTLY FOUR TIMES

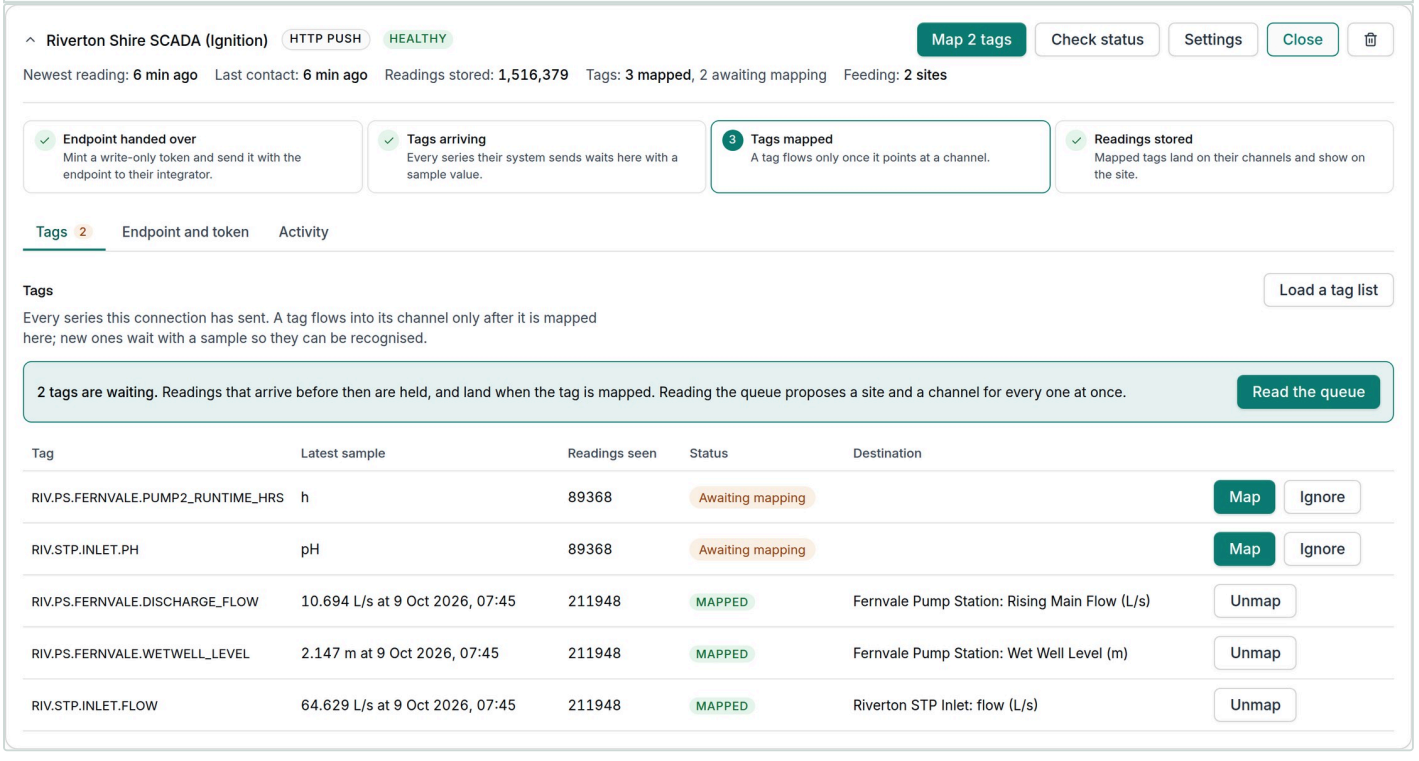
THE TAG YOUR SYSTEM SENDS	WHAT THE PLATFORM READS IT AS	WHY
`PS2_Pump1_Run`	Pump station 2, pump 1 run status	A station identifier, a pump number and a run contact.
`Riverton/PS02/Pmp1/RunSts`	The same thing, from an MQTT topic	The namespace every tag shares is dropped, but never a segment shaped like a station.

THE TAG YOUR SYSTEM SENDS	WHAT THE PLATFORM READS IT AS	WHY
`P1RUN`	The same thing again	Run together, split on the boundary between letters and digits.
`STP_Inlet_Flow.PV`	Flow at the inlet, with the historian's item suffix removed	`.PV` , `.CV` and `.STS` are historian furniture, not part of the name.
`PS2_FlowTotal` in kL	A meter index, differenced into a flow in L/s	A counter is read before the rate it totals, because a totaliser stored as it stands is a chart that climbs forever.
`PS4_HighLevelAlarm`	A high level alarm, not a level	Contacts are tested before hydraulics, or every alarm containing the word "level" becomes a level channel.

Proposals, never decisions

Reading the queue writes nothing. It answers "what would this do", station by station, including what it would reuse rather than create and what it refuses outright, so it is safe to run while somebody reads it out on a call. Applying it writes exactly what the operator confirmed and nothing it has inferred since, so a plan read an hour ago cannot quietly become a different plan.

Two refusals are deliberate. A counter whose unit cannot be converted into its channel's unit, a kilolitre index pointed at a channel in metres, is refused in front of the person configuring it rather than guessed at three in the morning. And one tag may never feed two channels, nor two tags one channel: a channel is one series, and that is what makes the analyses downstream mean anything.



A live connection, mid setup. The setup steps say where the connection has got to, and the tag table is the gate: three tags are mapped and flowing, two are waiting for somebody to recognise them, and their readings are held until they are pointed at a channel, then land with the mapping. Reading the queue proposes a site and a channel for every waiting tag at once.

5. WHAT USUALLY GOES WRONG, AND WHAT THE PLATFORM DOES ABOUT IT

Every line in this table was a defect somebody's data hit first.

This is the part of a SCADA integration that is hard to scope, because it is invisible until the data arrives. The behaviours below are what the platform does today, and are worth reading before your integrator plans around them.

The historian sends "Bad", "N/A" or a row of hashes instead of a number	Skipped and counted as skipped, never stored as zero and never counted as a rejection. A fault is not a measurement, and a zero in a flow series is.

A timestamp is local wall time in an hour that daylight saving deleted or repeated	Rejected with the reason, for connections in Sydney, Melbourne, Adelaide and Hobart, once a year. Brisbane, Perth and Darwin are unaffected. A time that means two different instants cannot be stored as either.
A reading is dated in the future	Rejected, with a five minute tolerance for ordinary clock drift. A clock set wrong at a site otherwise writes data across the top of a chart nobody can then read.
Your best instrument is a totaliser on the rising main	Declared as a counter and differenced on the way in. The first reading is a baseline, a backwards step re-bases unless you declare the rollover, and a gap longer than twenty five hours re-bases, because a guessed wrap invents the volume that went with it.
Two of your systems discover the same new tag at the same instant	One of them wins, the other reloads the winner's row, and both batches land. An earlier version refused the whole batch instead, which is why this exact race now has a regression test of its own.
The integrator re-sends last night's window "just to be sure"	Duplicates are dropped on the channel and timestamp. Re-sending is the supported recovery procedure, not an accident to be apologised for.
A tag arrives before anyone has mapped it	Held for seven days, up to five thousand readings for that tag, and replayed through the same validation the moment it is mapped. Mapping late costs you nothing but the delay.
A connection publishes seven of its eight tags	Each tag carries its own expected cadence, learned from its history, so the one that went quiet is flagged even though the connection is green by every other measure.
The file server lists files but every one is skipped	A real fault, found on ProFTPD and fixed: the transfer mode was being set before the listing, and the listing reset it, so every file size came back as an error and every file looked like a directory. The fix carries a regression test of its own.
Your broker or file server is unreachable for a day, or our folder is when your upload runs	Polls and pickups resume from where they stopped, and MQTT readings published at QoS 1 are queued by your broker against our persistent session and replayed, so nothing needs to be re-exported by hand. A file drop that cannot reach us fails at your end at the time, where your task can retry it; the file is still on your side, and sending it again later is safe.
Your team says the file went last night, and no readings arrived	For a file drop, our server's own record of every transfer answers it: complete, cut off part way, or refused with the reason, whether or not the file then turned into readings. A file that never arrived and a file that arrived empty stop looking the same.

6. FOR THE FIREWALL CHANGE REQUEST

Every connection, its direction and its port.

The platform is reached at `edsflowsense.au`. Where we connect to you, we will send the source address to allowlist before anything is configured, and confirm your server's fingerprint out of band before a credential is used. Where you send files to us, the login comes with our own server's fingerprint, for your client to pin.

CONNECTION	INITIATED BY	PROTOCOL AND PORT	AT YOUR END
Your staff using the platform	Their browser	HTTPS, TCP 443 to `edsflowsense.au`	Standard outbound web access. Plain HTTP is redirected to HTTPS.
HTTPS push of readings	Your SCADA	HTTPS, TCP 443 outbound	Nothing inbound. Allow outbound from the SCADA or historian host.
MQTT subscription	FlowSense	MQTT, TCP 8883 with TLS, or 1883	Broker reachable from our source address, and a broker user limited to the shared topics.
API poll	FlowSense	HTTPS, TCP 443	A public HTTPS endpoint. Plain HTTP, private addresses and redirects are refused.

CONNECTION	INITIATED BY	PROTOCOL AND PORT	AT YOUR END
File pickup	FlowSense	SFTP, TCP 22, or FTPS, TCP 21 with passive data ports	Server reachable from our source address, and an account confined to the export folder.
File drop	Your scheduled task	SFTP, TCP 2222 outbound	Nothing inbound. Allow outbound on 2222 from the host that sends the file. Not port 22: that is our own administration and no client service runs on it.
Scheduled feeds back to your systems	FlowSense	SFTP 22, FTPS 21, or MQTT 1883 and 8883	The same as pickup, in the other direction. Files are written under a temporary name and renamed on completion, so a watcher never reads a half written file.
Alarm webhooks	FlowSense	HTTPS, TCP 443	A public HTTPS endpoint of yours. Every request is signed so you can verify it came from us.

What we will never ask you for

- **OPC UA, DNP3, Modbus, EtherNet/IP or S7 exposed to the internet.** If a connection attempt lands on one of those ports, the platform names the protocol in its own error and stops.
- **A VPN, a jump host or any route into your control network.**
- **A login on your SCADA**, a directory account, or software installed on your servers or at your sites.
- **Write access to anything of yours.** Every credential you give us needs to read one thing.

7. WHAT ACTUALLY HAPPENS, IN ORDER

Typically an afternoon of your time, most of it waiting on a change request.

01 **Your team receives a setup form link**

Sent from inside the platform by whoever administers your account. It opens a form only: no login, no account, no access to any data. It can be passed between your engineering and IT teams as often as needed, and it expires after 45 days. If your policy keeps credentials out of web forms, leave those fields blank, say so, and we will arrange another way.

02 **You describe the connection, not your network**

Which of the five options, the hostname or endpoint, the port, whether TLS is used, the topics, folder or column headings, the time zone your timestamps are in, and the tag names each site reports under. The time zone is chosen from a list rather than typed, because **AEST** typed into a free text field can read every timestamp ten hours out.

03 **We send the source address, you make the firewall change**

Only for the three options where we connect to you. An HTTPS push needs no change at all, and a file drop at most an outbound rule for one port, so ask your SCADA team first whether they can push or send a file.

04 **An EDS engineer applies the form**

Submitting it creates nothing. An engineer reviews what you sent and creates the connection switched off, with every tag queued for mapping. Nothing has reached a chart yet.

05 **Tags are mapped with your people**

Each tag is matched to a channel with whoever knows what it measures, working from the platform's proposals. Until a tag is mapped its readings are counted and held, so a mistyped tag surfaces in a review queue, not as data.

06 **First readings, then switch on**

For a push, your integrator calls the identity endpoint, compares our clock with theirs and sends a batch; the answer says what happened to every row. For a file drop, your task sends one file and a test reads it without storing anything. For the other three, a test button runs the real connection once and reports what it found. Once the first readings land cleanly the connection is enabled, and from then on it is ordinary data.

8. CHECKLIST

Everything IT is asked for, on one page.

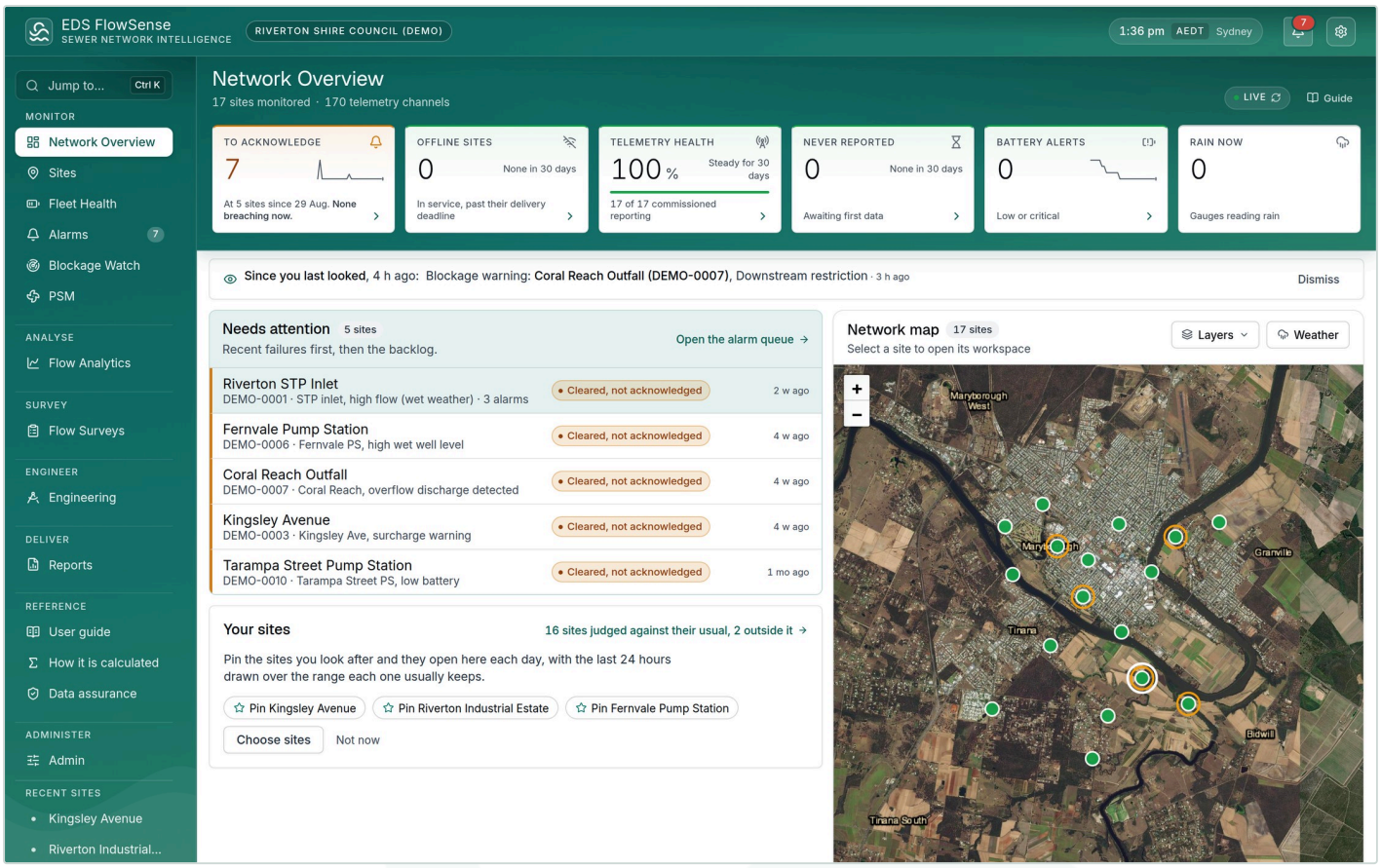
- [ ] Allow staff to reach `https://edsflowsense.au` on **TCP 443 outbound**, from any current version of Chrome, Edge, Firefox or Safari.
- [ ] Allow platform email to arrive: it is sent from the `edsflowsense.au` domain and delivered through `mg.edsflowsense.au`.
- [ ] Decide, with your control systems people, which of the five options your systems will use.
- [ ] **For an HTTPS push, nothing further.** Hand the endpoint and the write-only token to whoever writes the script.
- [ ] **For a file drop, one outbound rule.** Allow **TCP 2222 outbound** from the host that sends the file, and give whoever schedules the export the login and our host key fingerprint to pin.
- [ ] For the other three, allow our source address to reach **one** service, and create a read-only account scoped to one folder, topic tree or API.
- [ ] Confirm your server's certificate or host key fingerprint with us out of band, so it can be pinned.
- [ ] Approve the export of stored readings, if history is being loaded as well as live data.
- [ ] Nominate who administers your account, and decide whether two-factor authentication is mandatory for them.

9. WHERE THE READINGS LAND

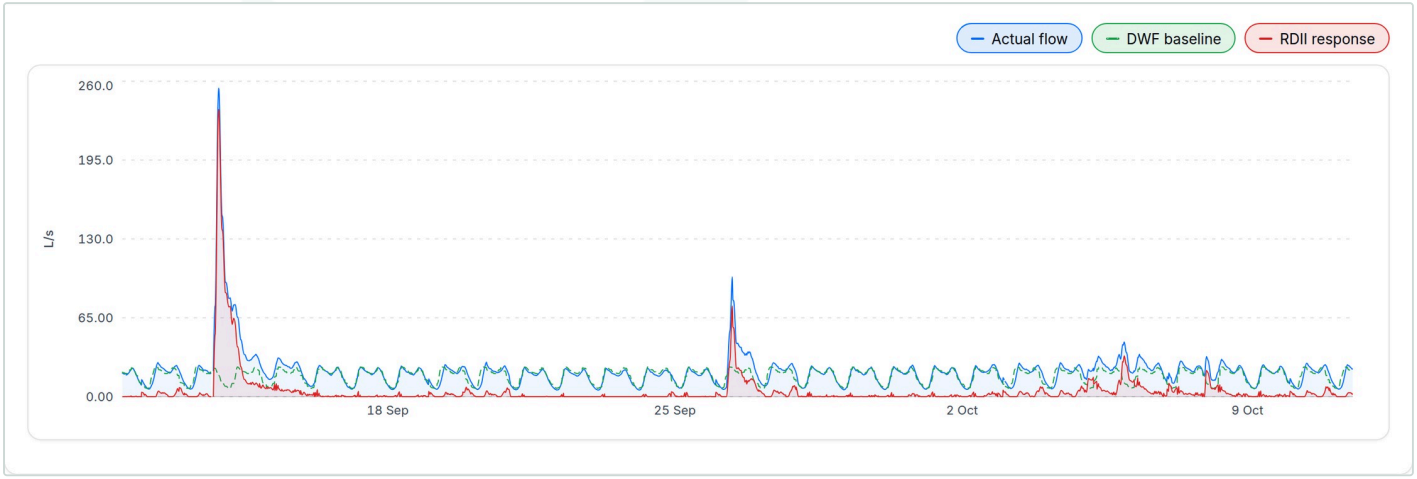
One platform, from the first reading to the report that goes to the regulator.

This is the argument for connecting at all. Most councils can already see a number on a SCADA screen. What they cannot do from a historian is tell how much of the flow is rainfall, which catchment it enters through, what the network will do in the next storm, and which pipe to send a crew to first. FlowSense is built for those questions, and it answers them on the data your SCADA is already collecting.

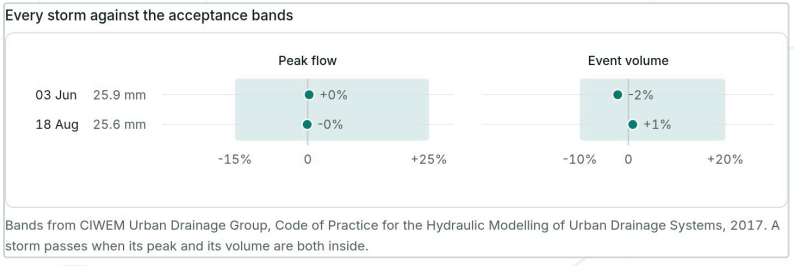
Monitor	Network overview, every site and channel, alarms on thresholds, on silence and on a pump's falling output with email and SMS to the people each rule names, blockage watch, pump station manager, fleet health with a replace-by date for every logger battery and, as a separate module, trade waste monitoring. Alarms are edge triggered with a cooldown, so a level hovering on a threshold does not send forty messages.
Analyse	Flow analytics: dry weather flow, the split between sewage, groundwater infiltration and rainfall derived inflow, storm by storm rainfall response, scattergraphs against the pipe's theoretical curve, and flow balances between meter pairs aligned on travel time. Rainfall comes from your own gauges where you have them, and from the ECMWF IFS weather model at each site's own grid cell where you do not.
Engineer	A modelling workstation that runs the chain in order: review the data, detect storms, fit the dry weather baseline, calibrate RTK unit hydrographs, validate the fit against storms held back, then put the calibrated response to work in a dynamic wave hydraulic model, a catchment routing model, pump station analysis and overflow forecasting. Built to the CIWEM Urban Drainage Group Code of Practice, WEF MOP FD-6, WSA 02, ARR 2019 and US EPA SWMM, with the scorecard the Code of Practice asks for.
Deliver	Seven report templates, generated on your data and branded with your logo, scheduled to arrive by email daily, weekly or monthly. Data export as CSV, through a read API, as scheduled files to your own server, and as SWMM, ICM, GeoJSON and Excel from the engineering tools. There is no format only EDS can read. The readings remain yours and can leave at any time.



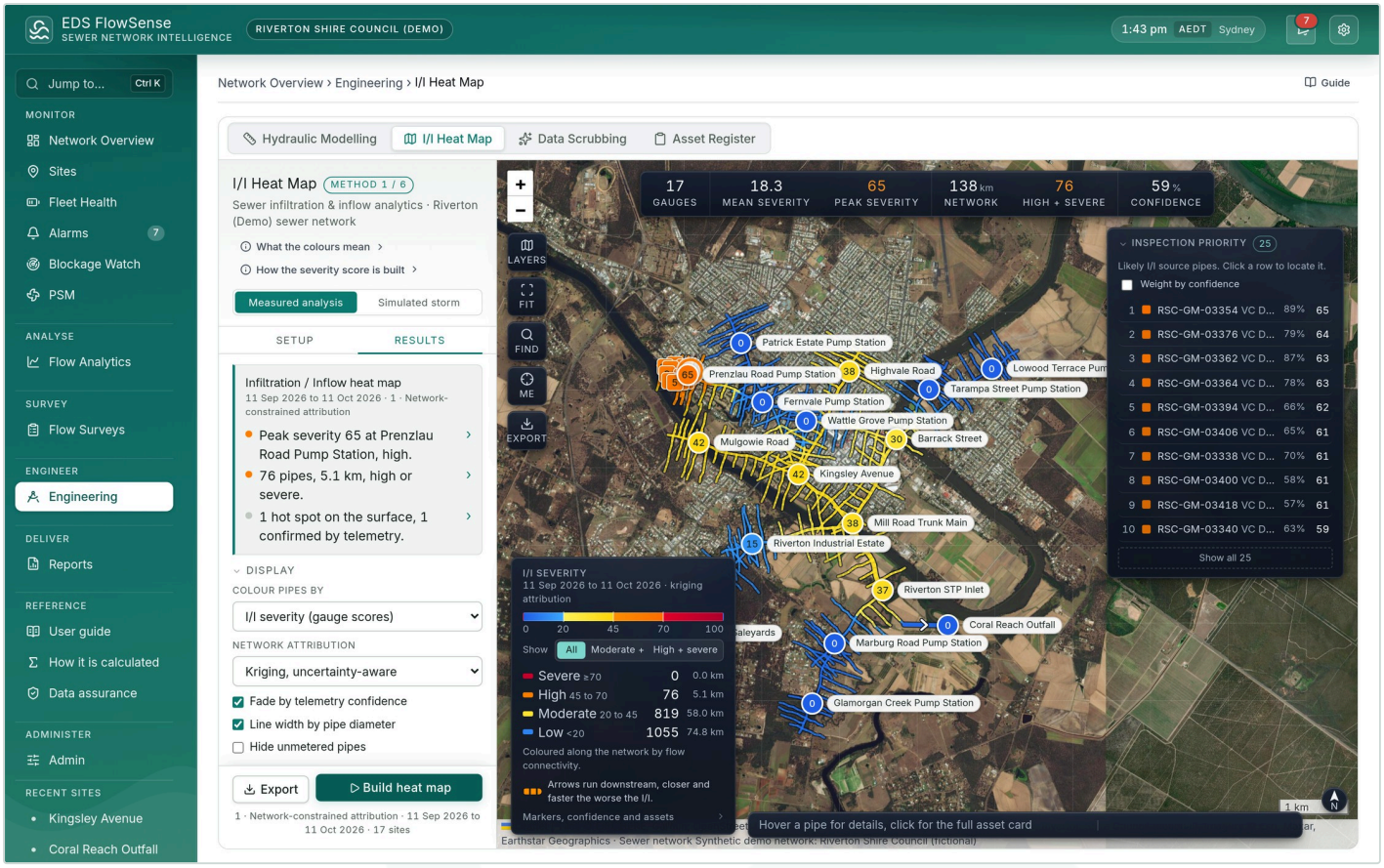
The operating picture. What needs attention sits beside where it is: each site waiting on somebody, newest first, saying whether its alarm is still breaching or has cleared and waits to be acknowledged, and pointing at a row rings that site on the map. The strip above counts alarms to acknowledge, sites past their own delivery deadline and loggers low on battery, and measures telemetry health against what each site was commissioned to deliver, so a site that has quietly stopped reporting is a number on this screen rather than a gap somebody notices next quarter.



What your flow meter was actually measuring. Thirty days at one catchment: the measured flow, the dry weather baseline the platform fits under it, and the difference between them, which is the rainfall that should never have been in the sewer. The baseline is a time varying groundwater term rather than a flat line, because infiltration rises and falls with the water table and a flat baseline charges that to the storm.



Calibration that scores itself. Each storm the model was fitted to is scored against the peak and volume acceptance bands published in the CIWEM Urban Drainage Group Code of Practice, and a storm passes only when both are inside. A single averaged score can look respectable while individual storms fail; these cannot. The errors are reported as signed numbers, so a model that runs consistently low is visible as such rather than averaged away, and the fit is then validated against storms held back from it.



Where the water gets in. Severity is interpolated across the gravity network by catchment and flow direction rather than by straight line distance, so a pipe on the other side of a ridge is not coloured by a gauge it does not drain to. Six analysis methods are selectable, the default attribution interpolates by kriging and carries a confidence surface, and the output is a ranked list of pipe identifiers a crew can be sent to, exportable as CSV.

Why this is better than the historian you already have

A historian stores. It answers "what was the level at 3am on Tuesday", and it answers it well. It does not know that 62 per cent of Tuesday's flow was not sewage, that the inflow entered through one branch of the catchment, that the response to the same rainfall has grown 40 per cent since the same month last year, or that the pump drawing more current at less flow has a blockage forming. Those are hydraulic questions, and they need a hydraulic model of your network, a calibrated rainfall response and a record of what the pipe is capable of, next to the telemetry rather than exported away from it.

The alternative most councils run is a historian, a consultant's spreadsheet and a modelling package on one engineer's desktop, with the analysis re-done from scratch every time somebody asks. FlowSense is the three of them in one place, on live data, with every step recorded and repeatable. Your historian keeps doing what it is good at, and the readings it collects do a second job on the way past.

Every calculated number carries a note saying how it was derived, and the platform holds a reference section written for the engineer who has to defend the figure in a meeting. A number nobody can account for is not worth the connection that produced it.



The document at the end of it. Generated on your data, branded with your logo, and scheduled to arrive by email. The executive summary states the period it covers and sets it against the previous period, the same period last year and rolling windows, so a figure is never presented without the context that makes it readable.

10. QUICK ANSWERS

The questions that come back from a review, answered in a sentence each.

Does anything have to be installed on our network?	No. Browser access, and at most a script or driver configuration on your own SCADA that your team controls.
Do you need inbound access to our control network?	No. An HTTPS push and a file drop are entirely outbound from you. For the other three we connect to one service you choose to expose, from a fixed source address you allowlist, and never to a plant protocol.
How long does a connection take to set up?	An HTTPS push can be sending the same day the token is issued, and a file drop the same day its outbound rule is in place. The others take as long as your change process takes, plus about an hour of configuration and tag mapping.
What if our tags are named inconsistently?	That is the normal case. The platform proposes a station and a channel for each tag from the name, and an operator confirms them in one pass rather than one at a time.
Can our SCADA create or change anything in FlowSense?	No. It can submit readings for tags a person has already mapped. An unknown tag is held for review and never creates a channel.
Can FlowSense change anything in our SCADA?	No. Every connection we make to your systems is read-only, and we ask for read-only credentials.
Can a leaked ingest token read our data?	No. Ingest tokens are a separate write-only credential class, tied to one connection, stored only as a hash and shown once.
Are our credentials encrypted at rest?	Yes, with authenticated encryption, and they are never returned by any interface after entry, to you or to us.

Is the data encrypted in transit?	Yes. TLS on every browser and API connection, and TLS or SSH on every connection the platform makes to your systems, with hostname checking never switched off.
What happens to readings if your platform is down?	Push retries and re-sent files are safe because duplicates are dropped. QoS 1 MQTT messages are queued by your broker and replayed. Polls and pickups resume from where they stopped. A file drop that cannot reach us fails at your end, where your task can retry it.
Do you alter our readings?	No. What is stored is what the instrument sent. A stored value changes only through a repair a person applies, and the value it replaced is recorded.
Can we get our data out again?	Yes, at any time, as CSV, through the read API, as scheduled files to your own server, and as standard engineering formats.
Is multi-factor authentication available?	Yes, time based one-time passcodes with recovery codes. EDS can require it for any role, and each person enrols at their next sign-in.
Can our security team review the controls?	Yes. They are set out in full in the technical and security brief, and a client security review is welcome.
Who do we talk to?	1300 721 683, and ask for the engineer looking after your organisation, or reply to the email this arrived with.

Travels with this	The **technical and security brief** answers a security questionnaire in full, the **integration guide** is the developer reference for the push endpoint, the read API and signed webhooks with working verification code, and the **historical data export specification** covers loading the history you already hold. Going the other way, the **outbound data brochure** sets out the four ways readings leave FlowSense again, into your SCADA, your historian or your reporting tools.
Before the connection	Loading exported history is a separate, earlier step needing nothing from IT but a browser and approval to send the files. It is worth doing first: your staff are then working on real data while the live connection goes through your change process.

NEXT STEP

Put us in front of the person who would build it.

An EDS engineer will walk your control systems or IT team through the five options on a call, tell you which is least work for what you already run, send the source address for any firewall change, and confirm the peer fingerprint out of band before a credential is used. If the answer turns out to be an HTTPS push or a file drop, that call is usually the whole project meeting.

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EDS FlowSense SCADA connection brochure, October 2026. Every endpoint, limit and behaviour described is read from the released platform at the date of issue.

DOCUMENT CONTROL

DOCUMENT	REFERENCE	REVISION
EDS FlowSense SCADA connection brochure	EDS-FS-SCA-01	**2.5**, issued 12 October 2026
SUPERSEDES	**STATUS**	**ISSUED BY**
Revision 2.4, 11 October 2026	Issued for client use	Environmental Data Services, ABN 31 729 813 151